

Skills Summary

Choosing and Combining Factoring Methods

Use this reference while completing your worksheet or when studying.

Run this checklist in order on every expression.

- **1. Common factor?** Pull out the GCF of all terms, variables included. Then look at the bracket and start again from step 2.
- **2. How many terms?** Two terms with a minus sign: difference of squares, $A^2 - B^2 = (A - B)(A + B)$. Three terms: trinomial.
- **3. Perfect square?** $A^2 \pm 2AB + B^2 = (A \pm B)^2$ — square ends, middle term equal to $2AB$.
- **4. Leading coefficient 1 or not?** If 1, find a factor pair of the constant adding to the middle coefficient. If not, group on the product of the outer coefficients.
- **5. Factor again.** Every bracket gets the checklist too. A sum of squares, $A^2 + B^2$, is where you stop.

1. Common factor first, then look again

The GCF step is not optional politeness — it is what makes the later pattern visible. A coefficient that is not a perfect square ($2x^2$, $3x^2$) or a term of degree three or higher is the usual sign that a common factor is hiding a pattern underneath.

Example: Factor $2x^3 - 8x$ completely.

Step 1. Two terms and a minus sign look like a difference of squares, but $2x^3$ is not a square — so run checklist step 1 first and take the common factor out.

Step 2. Both terms divide by 2 and both carry an x , so the GCF is $2x$: $2x^3 - 8x = 2x(x^2 - 4)$, and $x^2 - 4 = (x - 2)(x + 2)$. $2x^3 - 8x = 2x(x - 2)(x + 2)$

Try it: Factor $3x^3 - 27x$ completely.

$(x + x)(x - x)x$:check

2. Trinomials with leading coefficient 1

$x^2 + bx + c = (x + p)(x + q)$ where $p + q = b$ and $pq = c$. Read the signs before hunting: if $c > 0$ both numbers share the sign of b ; if $c < 0$ they have opposite signs, and the bigger one carries the sign of b .

Example: Factor $x^2 + 7x + 12$.

Step 1. No common factor and the leading coefficient is 1, so this is the factor-pair case — no grouping needed.

Step 2. Need a pair multiplying to 12 and adding to 7: 3 and 4, since $3 \cdot 4 = 12$ and $3 + 4 = 7$. $x^2 + 7x + 12 = (x + 3)(x + 4)$

Try it: Factor $x^2 - 2x - 15$.

$(x + 5)(x - 3)$:check

3. Trinomials with a leading coefficient that is not 1

For $ax^2 + bx + c$ with $a \neq 1$: find a factor pair of the product ac that adds to b , split the middle term into those two pieces, then factor by grouping in pairs. The two brackets you get must match, which is the built-in check that the split was right.

Example: Factor $3x^2 + 11x + 6$.

Step 1. No common factor and the leading coefficient is 3, so factor pairs of the constant alone will not work — group on the product ac instead.

Step 2. $ac = 3 \cdot 6 = 18$, and $9 + 2 = 11$, so split: $3x^2 + 9x + 2x + 6 = 3x(x + 3) + 2(x + 3)$.
 $3x^2 + 11x + 6 = (3x + 2)(x + 3)$

Try it: Factor $2x^2 - 5x - 3$.

$(2x + 1)(x - 3)$:check

4. Special forms: squares you can spot

$A^2 - B^2 = (A - B)(A + B)$ and $A^2 \pm 2AB + B^2 = (A \pm B)^2$. Squares to know on sight: 4, 9, 16, 25, 36, 49, 64, 81, 100; also $4x^2 = (2x)^2$, $9x^2 = (3x)^2$, $x^4 = (x^2)^2$. Recognising a special form skips all trial and error.

Example: Factor $4x^2 - 20x + 25$.

Step 1. Three terms whose first and last are perfect squares — test the perfect-square pattern before grouping, because it is far quicker when it works.

Step 2. $A = 2x$ and $B = 5$; the pattern needs a middle term of $2AB = 2 \cdot 2x \cdot 5 = 20x$, and the trinomial has $-20x$, so the bracket is a difference. $4x^2 - 20x + 25 = (2x - 5)^2$

Try it: Factor $9x^2 - 24x + 16$.

$(3x - 4)^2$:check

(!) Watch out: A correct product is not always a complete factorization. $(2x + 4)(x + 2)$ really does multiply back to $2x^2 + 8x + 8$, but $2x + 4$ still hides a common factor of 2 — the complete answer is $2(x + 2)^2$. Taking the GCF out first would have prevented it.